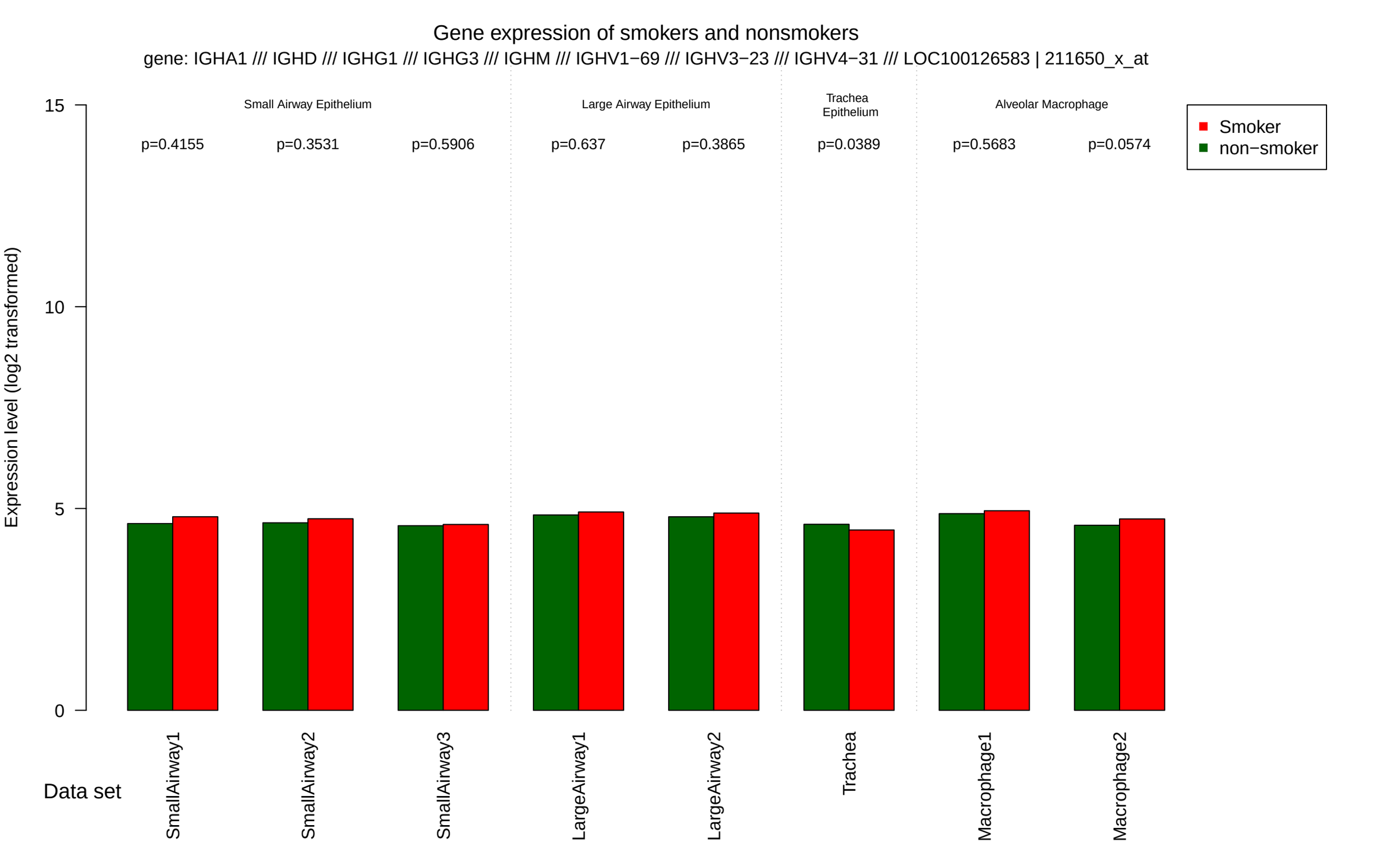
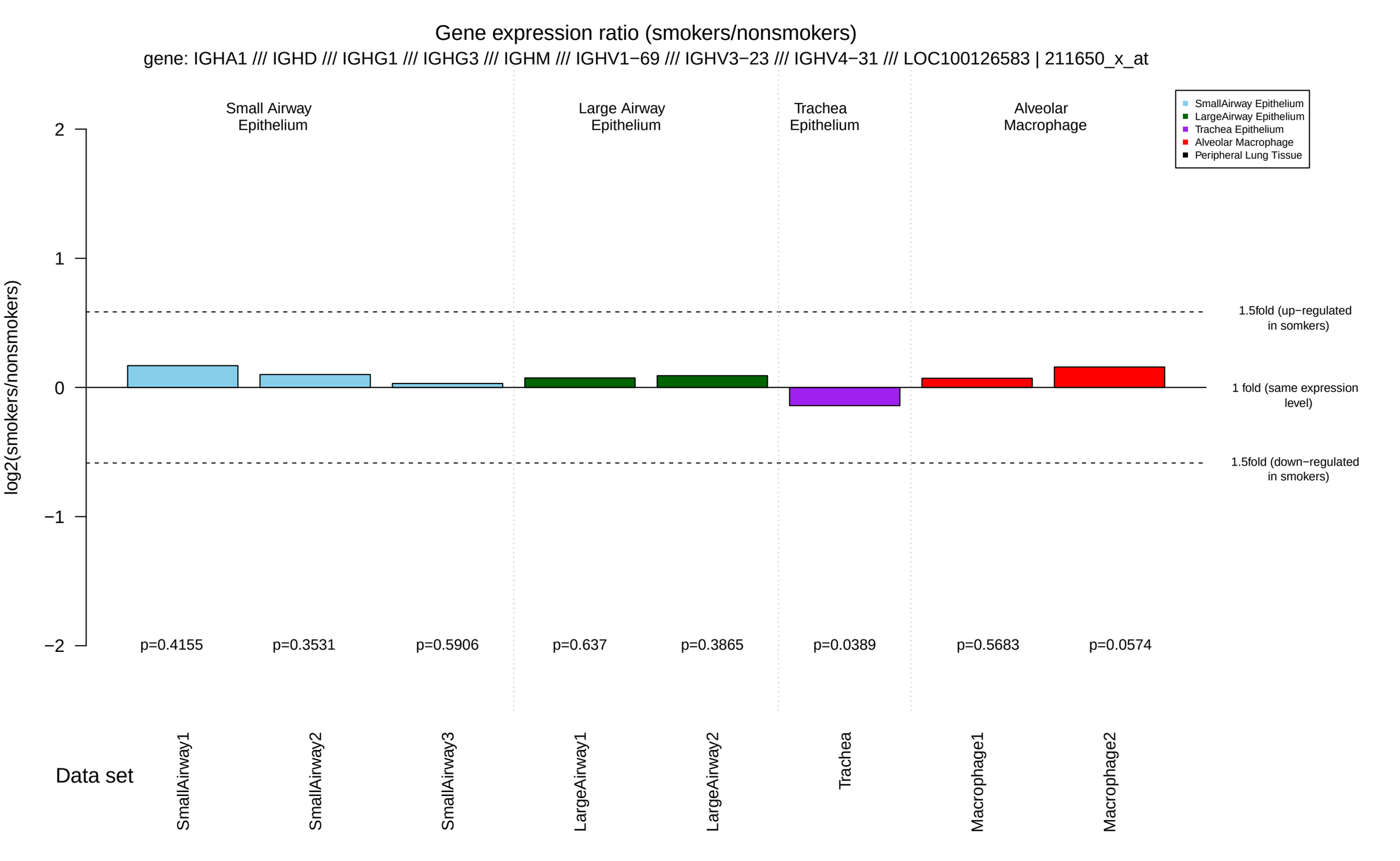
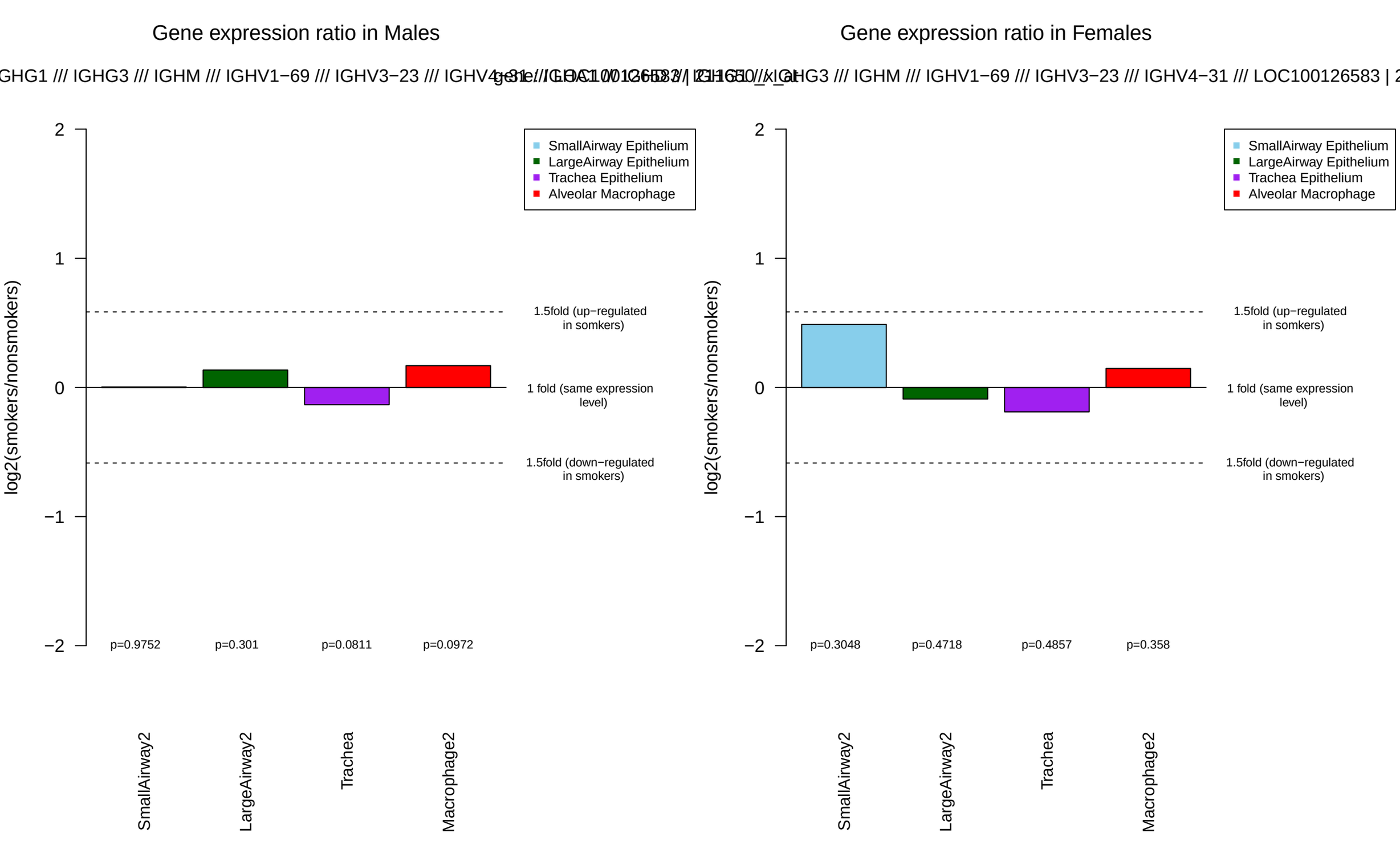
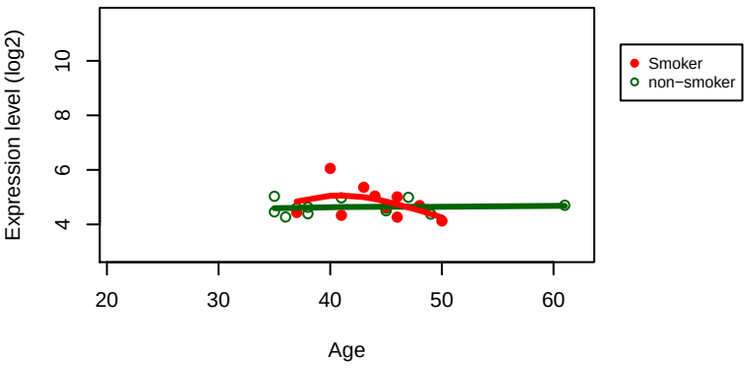

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV1−69 /// IGHV3−23 /// IGHV4−31 /// LOC100126583 | 211650_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4155
p=0.3531
p=0.5906
p=0.637
p=0.3865
p=0.0389
p=0.5683
p=0.0574
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

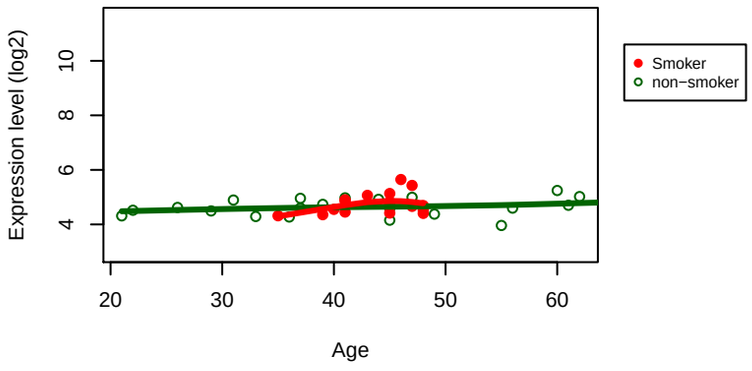

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV1−69 /// IGHV3−23 /// IGHV4−31 /// LOC100126583 | 211650_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4155
p=0.3531
p=0.5906
p=0.637
p=0.3865
p=0.0389
p=0.5683
p=0.0574
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio in Males
IGHG1 /// IGHG3 /// IGHM /// IGHV1−69 /// IGHV3−23 /// IGHV4−31 /// LOC100126583 | 23163_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9752
p=0.301
p=0.0811
p=0.0972
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
IGHG1 /// IGHG3 /// IGHM /// IGHV1−69 /// IGHV3−23 /// IGHV4−31 /// LOC100126583 | 23163_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3048
p=0.4718
p=0.4857
p=0.358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

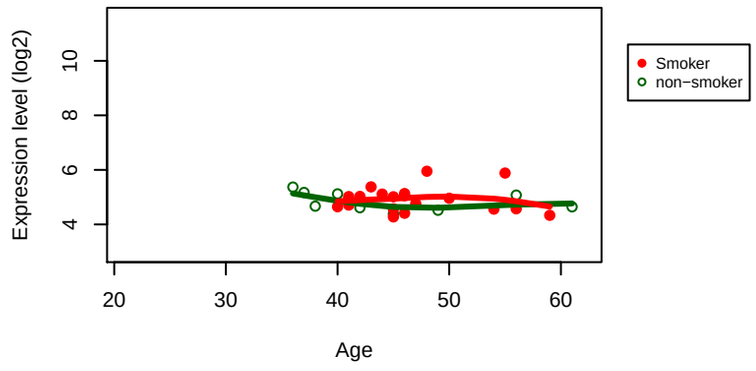
set: SmallAirway1 (p=0.4155)



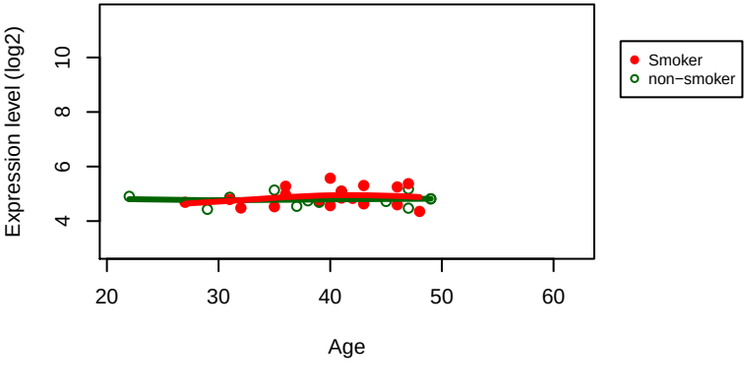
set: SmallAirway2 (p=0.3531)



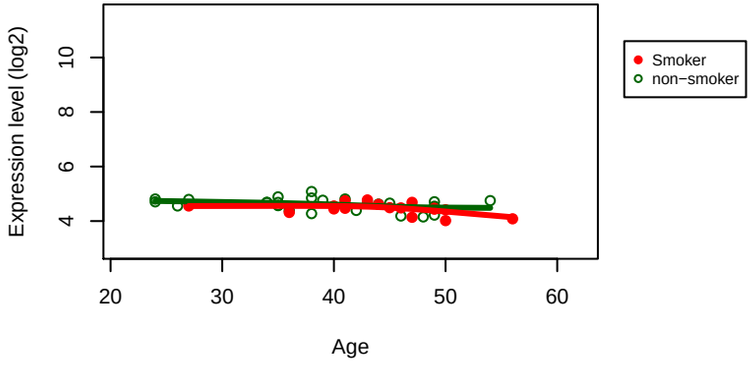
set: LargeAirway1 (p=0.637)



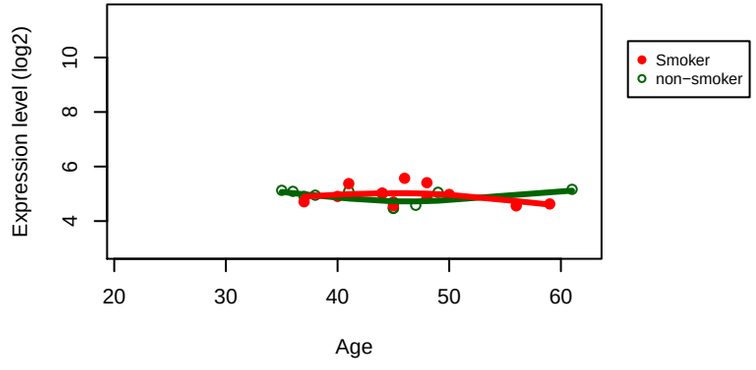
set: LargeAirway2 (p=0.3865)



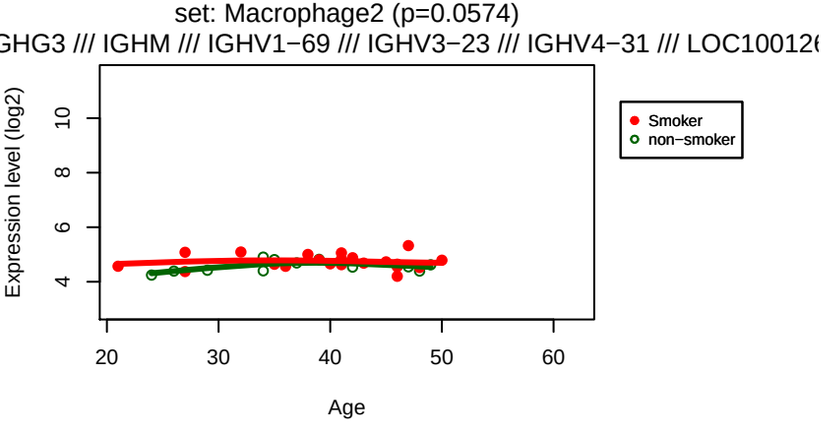
set: Trachea (p=0.0389)

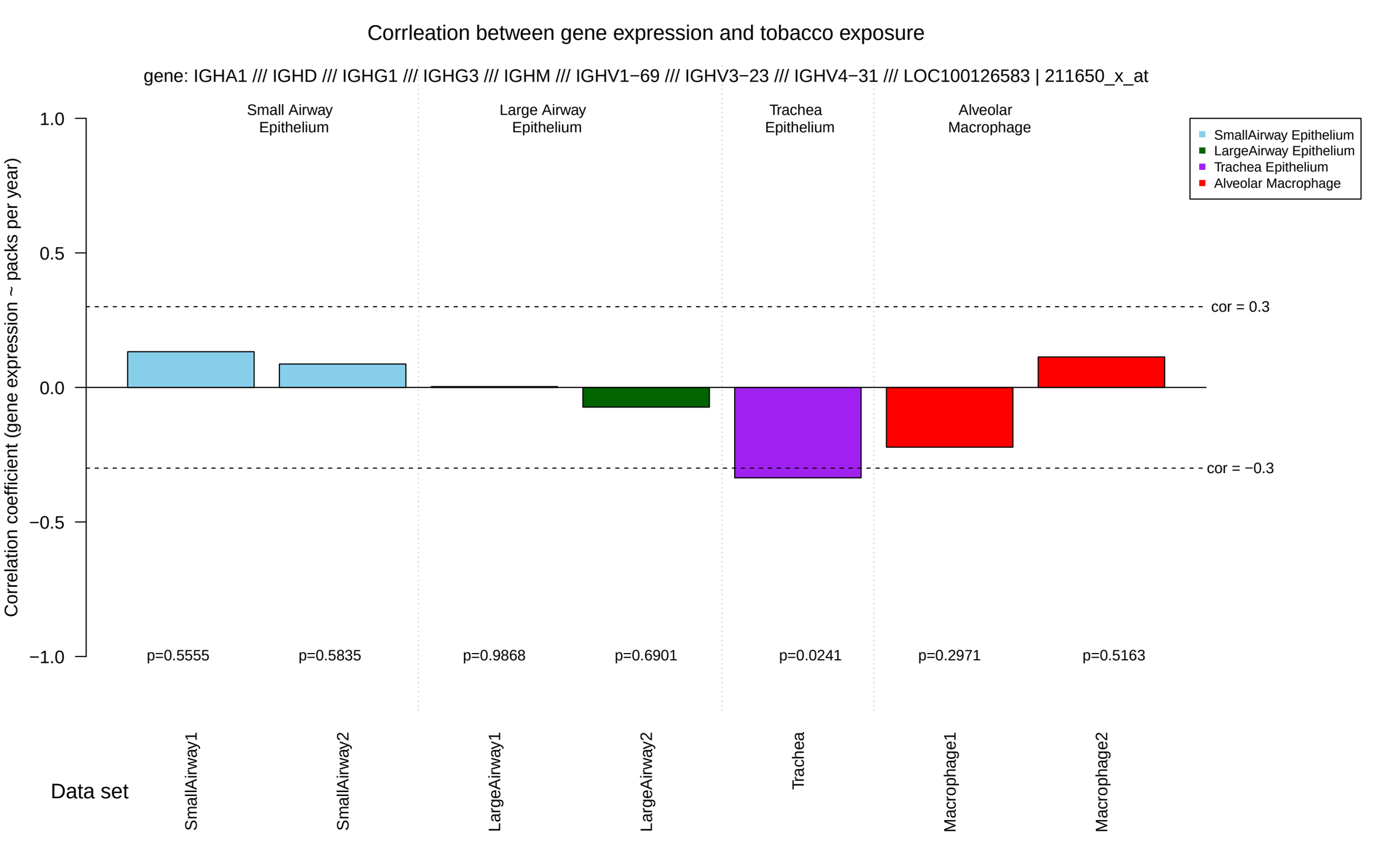


set: Macrophage1 (p=0.5683)



set: Macrophage2 (p=0.0574)




Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV1−69 /// IGHV3−23 /// IGHV4−31 /// LOC100126583 | 211650_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5555
p=0.5835
p=0.9868
p=0.6901
p=0.0241
p=0.2971
p=0.5163
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2